UNIT III

Helical, Bevel and Worm Gears

parallel helical | crossed helical | straight bevel | worm gearing | kinematics | forces | stresses | friction & efficiency | design

I. Parallel helical gears

Helical gears have teeth cut at a **helix angle β** to the axis. Contact starts at one end of the tooth and spreads gradually along an inclined line, so the load is picked up smoothly - **quieter, stronger, higher speed** (up to 50 m/s) than spur gears, but they produce **axial thrust**. Used in automobile **constant-mesh gearbox**, differential final drives, timing drives.

Feature	Spur	Helical
Contact	Line across whole face at once	Gradual along inclined line; 2-3 teeth in contact
Noise / smoothness	Noisy at high speed	Quiet, smooth
Load capacity	Lower	Higher (longer contact line, higher contact ratio)
Axial thrust	None	Yes: $F_a = F_t \tan \beta$ (use thrust/taper bearings, or double helical)
Min. teeth (20° FD)	17	$17 \cos^3\beta$ (about 15-16 at β = 15-20°)
Cost	Cheap	Dearer (cutting with helical hob / cutters)

Kinematics and tooth proportions

Quantity	Formula
Hand of helix	Parallel shafts: pinion RH ↔ gear LH (opposite hands, same angle β)
Normal module / circular pitch	m_n ; $p_n = \pi m_n$ (standard values, cutter cuts normal section)
Transverse module / pitch	$m_t = m_n / \cos\beta$; $p_t = \pi m_t = p_n / \cos\beta$
Axial pitch	$p_x = p_t / \tan\beta = \pi m_n / \sin\beta$
Pressure angles	$\tan\varphi_t = \tan\varphi_n / \cos\beta$ (φ_n = 20°)
Pitch diameter	$d = m_t z = m_n z / \cos\beta$
Centre distance	$a = m_n (z1 + z2) / (2 \cos\beta)$ (adjustable by changing β)

Quantity	Formula
Virtual (formative) teeth	$z' = z / \cos^3\beta$ - the equivalent spur gear in the normal plane (used in Lewis equation)
Velocity ratio, speed	$i = z2/z1 = N1/N2$; $v = \pi\, d1\, N1 / 60000$
Face width	$b \geq px = \pi\, mn/\sin\beta$ (overlap) for helical action; usually $b = 10\ mn$ to $12.5\ mn$ or $b \geq 1.15\ px - 2\ px$
Helix angle	$\beta = 15°\text{-}30°$ (single helical); 30°-45° for double helical (herringbone)
Contact ratio	Total CR = transverse CR + axial CR (b/px) ; total ≥ 2 is very smooth

Force analysis on a helical gear tooth

The normal force Fn lies in the normal plane; resolve it into three perpendicular components at the pitch point:

Component	Formula	Effect
Tangential Ft	$Ft = 2T / d1 = 2000\ P / v$	Transmits power; bends tooth, twists shaft
Radial Fr	$Fr = Ft \tan\varphi n / \cos\beta = Ft \tan\varphi t$	Separating force; bends shaft
Axial (thrust) Fa	$Fa = Ft \tan\beta$	Thrust bearing load; bends shaft with moment Fa d/2
Normal force Fn	$Fn = Ft / (\cos\varphi n \cos\beta)$	Resultant

> Larger β -> smoother but larger thrust. Thrust direction depends on hand of helix and rotation; choose hand so that thrust loads the shaft bearing with the larger capacity.

> **Double helical (herringbone)** gears: two opposite helices on one blank; axial thrusts **cancel** inside the gear, so high β (up to 45°) can be used - heavy-duty reducers, turbines.

Stresses in helical gear tooth

$$\sigma b = Ft / (b\ mn\ Y') \quad \text{with} \quad Y' = \pi\, y', \quad y' = 0.154 - 0.912 / z' \quad (z' = z / \cos^3\beta)$$

> **Beam strength** $Sb = \sigma b\ mn\ b\ Y'$; dynamic load $Fd = Cs\ Ft / Cv$ ($Cv = 3/(3 + v)$, $4.5/(4.5 + v)$ or $6/(6 + v)$ as for spur gears).

> **Wear (Buckingham)** $Sw = b\ Q\ d1\ K / \cos^2\beta$ with $Q = 2\ z2/(z1 + z2)$, $K = 0.16\ (BHN/100)^2$.

> Check **Sb ≥ Fd** and **Sw ≥ Fd**; the weaker of pinion/gear ($\sigma b\ y'$) governs.

> Helical gears carry about **30–50% more power** than equal spur gears because of the longer line of

contact.

Design of helical gears - steps

1. Select β (15°-25° typical), $\varphi n = 20°$, material (σb, BHN), $z1$ ($\geq 17 \cos^3\beta$), $z2 = i\ z1$.
2. $T = 9.55 \times 10^6\ P\ Cs / N1$; assume $b = 10\ mn$; $z1' = z1/\cos^3\beta$ -> $y1'$ (and $y2'$).
3. $d1 = mn\ z1 / \cos\beta$; $v = \pi\ d1\ N1 / 60000$ -> Cv. Design the weaker gear for strength:

$$mn^3 = 2\ T \cos\beta\ /\ [\ z1\ \sigma b\ Cv\ (b/mn)\ \pi\ y'\ ] \quad \rightarrow \quad \text{standard } mn$$

4. Calculate d1, d2, $a = (d1 + d2)/2$, b; check $b \geq px$ (axial overlap).
5. $Fd = Ft / Cv$; check $Sb \geq Fd$ and wear $Sw = b\ Q\ d1\ K / \cos^2\beta \geq Fd$ -> required BHN.
6. Find axial thrust Fa, radial Fr and select bearings; check shaft design.

Example 1 - helical gear design

> P = 10 kW, N1 = 1000 rpm, i = 3, $\beta = 20°$, $\varphi n = 20°$, z1 = 20, z2 = 60, σb = 160 MPa (both), ordinary cut gears. T = 95,500 N·mm.

> $z1' = 20/\cos^3 20° = 24.1$ -> $y' = 0.154 - 0.912/24.1 = 0.1162$; $Y' = \pi\ y' = 0.365$. $d1 = mn \times 20/0.9397 = 21.28\ mn$; $v = 1.114\ mn$ (m/s).

> $mn^3\ Cv = 2T \cos\beta/[z1\ \sigma b\ (b/mn)\ \pi\ y'] = 2 \times 95{,}500 \times 0.9397/(20 \times 160 \times 10 \times 0.365) = 15.4$. mn = 3: v = 3.34, Cv = 0.473, $mn^3\ Cv = 12.8$ ✗; mn = 4: v = 4.46, Cv = 0.402, $mn^3\ Cv = 25.7$ ✓ -> **mn = 4 mm**.

> d1 = 85.1, d2 = 255.4, a = 170.3 mm; $b = 40\ mm \geq px = \pi \times 4/\sin 20° = 36.7$ ✓. $Ft = 2T/d1 = 2244$ N; $Fd = Ft/Cv = 5582$ N; $Sb = 160 \times 4 \times 40 \times 0.365 = 9344$ N ✓.

> Wear: Q = 1.5; need $K \geq Fd \cos^2\beta/(b\ Q\ d1) = 5582 \times 0.883/(40 \times 1.5 \times 85.1) = 0.965$ -> BHN ≥ 246 -> **BHN 250**. $Fa = Ft \tan 20° = 817$ N; $Fr = Ft \tan 20°/\cos 20° = 869$ N.

II. Crossed helical gears (screw gears)

Two helical gears mounted on **non-parallel, non-intersecting shafts**. They have **point contact** (the teeth are tangent at a point), so the load capacity is **low** and sliding is high. Used for **light drives** - oil pumps, distributor / speedometer drives in automobiles, instruments.

Item	Relation
Shaft angle	$\Sigma = \beta1 + \beta2$ (same hand) or $\beta1 - \beta2$ (opposite hands); 90° drive: $\beta1 + \beta2 = 90°$

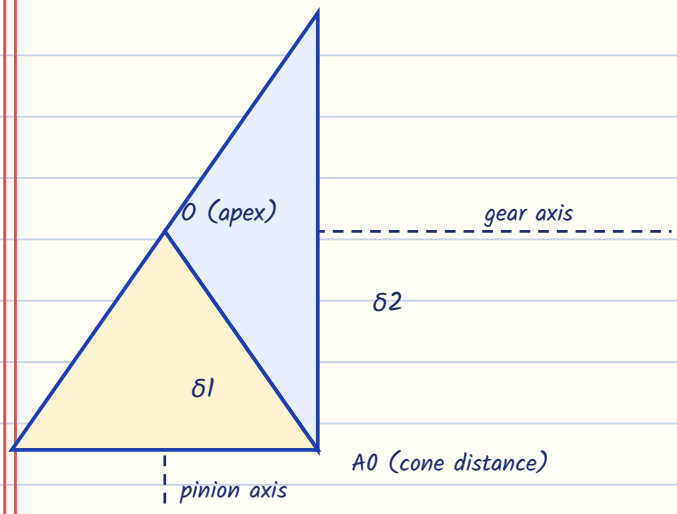
Item	Relation
Normal module	mn same for both gears
Pitch diameters	$d1 = mn\ z1 / \cos\beta1$; $d2 = mn\ z2 / \cos\beta2$
Centre distance	$a = (d1 + d2)/2 = (mn/2)(z1/\cos\beta1 + z2/\cos\beta2)$
Velocity ratio	$N1/N2 = z2/z1$ (NOT d2/d1 because β may differ); $= d2\cos\beta2 / (d1\cos\beta1)$
Efficiency	$\eta = (\cos\varphi n - \mu\tan\beta1) / (\cos\varphi n + \mu\tan\beta2)$ (for lubricated drives 90-95% at best helix angles)
Max efficiency	when $\beta1 = \beta2 = \Sigma/2$ (equal helix angles for 90° drive: 45° each)
Forces	Normal force Fn gives Ft1, Fa1 on gear 1 equal to Fa2 , Ft2 on gear 2: Ft1 = Fa2 and Fa1 = Ft2 (for 90°), plus friction component

> Design: beam strength by Lewis on virtual teeth; the limit is **wear**: permissible load Fw = k d b (k from data book); low load so it is checked for wear first.

> Materials: hardened steel on bronze or steel on cast iron; lubricate well.

III. Straight bevel gears

Bevel gears connect **intersecting shafts** (usually 90°). Pitch surfaces are **cones**. **Straight bevel**: teeth are straight and point towards the apex; cheap, used for low speed (< 5 m/s); axial thrust small. **Spiral bevel**: curved oblique teeth - smooth, high speed. **Hypoid**: shafts offset, used in automobile **differential / final drive**. **Zerol**: curved with zero spiral angle.



$\Sigma = \delta1 + \delta2$ (shaft angle, 90°)

$\tan\delta1 = z1 / z2$; $\tan\delta2 = z2 / z1$

$A0 = d1 / (2\sin\delta1) = d2 / (2\sin\delta2)$

Face width $b \le A0 / 3$ (or 10 m)

Virtual teeth $z' = z / \cos\delta$

Forces act at the **mean diameter**

$dm = d - b\sin\delta$

Kinematics and geometry

Quantity	Formula
Pitch cone angles	tan δ1 = sin Σ/(z2/z1 + cos Σ); for Σ = 90°: tan δ1 = z1/z2 , tan δ2 = z2/z1
Velocity ratio	i = N1/N2 = z2/z1 = sin δ2 / sin δ1
Cone distance	A0 = d1/(2 sin δ1) = (m/2) √(z1² + z2²) (Σ = 90°)
Face width	b ≤ A0/3 and b ≤ 10 m (take smaller; more -> poor contact at small end)
Mean pitch dia	dm = d - b sin δ ; mean module mm = m (1 - b / (2 A0))
Back cone (Tredgold)	Virtual spur gear radius Rb = r / cos δ -> **virtual teeth z' = z / cos δ**
Pitch velocity	v = π d1 N1 / 60000 (at the large end)
Addendum / dedendum	ha = m, hf = 1.2 m (Gleason: ha = 1.0 m, total depth 2.188 m)

Force analysis (Σ = 90°)

Forces are assumed to act at the **mid-point of the face width** (mean diameter dm1 = d1 - b sin δ1).

Force	Pinion	Gear
Tangential	Ft = 2T1 / dm1	Ft (equal and opposite)
Radial	Fr1 = Ft tan φ cos δ1	Fr2 = Ft tan φ cos δ2 = Fa1
Axial	Fa1 = Ft tan φ sin δ1	Fa2 = Ft tan φ sin δ2 = Fr1

> Pinion axial thrust = gear radial load, and pinion radial = gear axial (for Σ = 90°) because the axes are perpendicular.

> Axial thrust pushes the gears **apart** - bearings must locate bevel gears precisely (taper roller bearings).

> Mount bevel gears close to a bearing (avoid overhang); the shafts must be rigid.

Stresses and design of straight bevel gears

$$S_b = \sigma_b\, m\, b\, Y' (1 - b / A_0) \quad ; \quad S_w = b\, Q\, d_1\, K / \cos\delta_1$$

> Lewis on virtual teeth z' = z / cos δ, y' = 0.154 - 0.912/z'. The factor (1 - b/A0) = (A0 - b)/A0 accounts for the tapering tooth thickness (so use b ≤ A0/3).

> **Dynamic load** Fd = Ft / Cv ; Cv = 6 / (6 + v) (v = pitch-line velocity, for cut/ generated bevel gears).

> **Ratio factor** Q = 2 z2' / (z1' + z2') with virtual teeth; K = 0.16 (BHN/100)².

> Design steps: (1) materials, z1 ≥ 17 (virtual teeth), z2; (2) δ1, δ2, T; (3) assume b = A0/3 ~ 10m; (4) find m from Sb ≥ Fd; (5) check wear, give BHN; (6) calculate forces & bearings.

Example 2 – straight bevel gear pair

> P = 5 kW, N1 = 1000 rpm, i = 2, Σ = 90°, φ = 20°, σb = 140 MPa, z1 = 20, z2 = 40. T = 47,750 N·mm. δ1 = $\tan^{-1}(1/2)$ = 26.57°, δ2 = 63.43°.

> A0 = 22.36 m ; b = A0/3 = 7.45 m ; dm1 = m z1 (1 - b/2A0) = 16.67 m -> Ft = 2T/dm1 = 5729/m. z1' = 20/cos26.57° = 22.36 -> y' = 0.1132, Y' = 0.3556.

> Sb = σb m b Y'(1 - b/A0) = 140 × 7.45 m² × 0.3556 × 0.667 = 247 m² ; Fd = Ft/Cv, v = 1.047 m. m = 3: Cv = 0.656 -> Fd = 2911 N > Sb = 2223 N ✗ ; m = 4: Cv = 0.589, Ft = 1432, Fd = 2431 N ; Sb = 247 × 16 = 3952 N ✓ -> **m = 4 mm**.

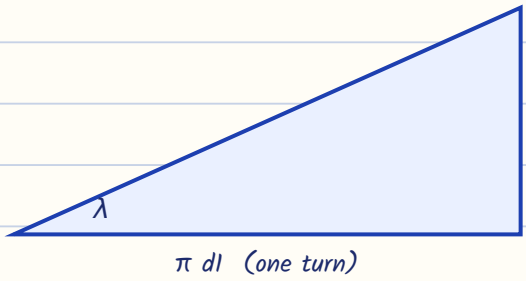
> d1 = 80, d2 = 160, A0 = 89.4, b = 30 mm. Wear: z2' = 89.4 -> Q = 1.6 ; K ≥ Fd cos δ1/(b Q d1) = 2431 × 0.894/(30 × 1.6 × 80) = 0.566 -> BHN ≥ 188. Forces: Fr1 = Fa2 = 1432 tan20° cos26.57° = 466 N ; Fa1 = Fr2 = 233 N.

IV. Worm gearing

A **worm gear drive** has a screw-like **worm** meshing with a **worm wheel**; shafts are **perpendicular, non-intersecting**. It gives a **high reduction ratio (5:1 to 100:1) in one stage**, quiet, compact, can be **self-locking**. Low efficiency (40-95%), heat generation, thrust. Uses: automobile **steering gear**, hoists, conveyors, lifts, speedometer, Torsen differential.

$\tan \lambda = L / (\pi d1)$

$= z1\ p / (\pi d1) = z1\ m / d1$



z1 = starts of worm (1, 2, 4)

q = d1 / m (diametral quotient)

i = z2 / z1

Term	Meaning / formula
Number of starts z1	Threads on the worm: 1, 2, 4 (or 6). Wheel teeth z2 ≥ 24 (30-60 typical)
Velocity ratio	i = z2 / z1 = N1 / N2
Axial pitch px	px = π m ; lead L = z1 px = π m z1
Lead angle λ	tan λ = L/(π d1) = z1 m / d1 = z1 / q

Term	Meaning / formula
Diametral quotient q	$q = d1 / m$ (8 to 17; 10 common); $d1 \approx 0.3\,a$ to $0.5\,a$
Pitch diameters	$d1 = q\,m$; $d2 = m\,z2$
Centre distance	$a = (d1 + d2)/2 = m\,(q + z2) / 2$
Face width of wheel	$b \approx 0.75\,d1$ ($z1 \le 3$); $0.67\,d1$ ($z1 > 3$); worm length $\approx 2.5\,m\,\sqrt{(z2 + 1)}$ (approx.)
Pressure angle	$\varphi n = 14.5°$ ($\lambda \le 15°$), $20°$ ($15°$-$30°$), $25°$ ($\lambda > 30°$) - bigger λ needs bigger angle

> Types: **cylindrical** (non-throated), **single-enveloping** (wheel throated - more contact), **double-enveloping / Cone (Hindley)** - highest capacity.

> **Materials**: worm = case-hardened & ground steel (20Cr/ 20MnCrS, 58-62 HRC); wheel = **phosphor bronze** / tin bronze (centrifugally cast), cast iron for light loads (bronze rim on cast-iron centre).

Forces on worm gearing and friction

Force	Without friction	With friction (coefficient μ)
Tangential on worm $Wt1$ = axial on wheel	$Wt1 = Wa1 \tan \lambda$	$Wt1 = Fn\,(\cos \varphi n \sin \lambda + \mu \cos \lambda)$
Axial on worm $Wa1$ = tangential on wheel $Wt2$	$Wa1 = 2\,T2 / d2$	$Wa1 = Fn\,(\cos \varphi n \cos \lambda - \mu \sin \lambda)$
Radial Wr (separating)	$Wr = Wa1 \tan \varphi n / \cos \lambda$	$Wr = Fn \sin \varphi n$

> Worm torque $T1 = Wt1\,d1/2$; wheel torque $T2 = Wa1\,d2/2$; note **$Wt2 = Wa1$, $Wa2 = Wt1$, $Wr2 = Wr1$**.

> **Sliding velocity** $Vs = v1 / \cos \lambda = \pi\,d1\,N1 / (60000 \cos \lambda)$ (m/s) - the whole tooth slides, causing heat.

> **Coefficient of friction** depends on Vs and materials: $\mu \approx 0.10$ at $Vs = 0.1$ m/s, 0.05 at 1 m/s, 0.03 at 5 m/s, 0.02 at ≥ 10 m/s (hard worm on bronze, oil-lubricated; use data-book curve).

Efficiency and self-locking

$$\eta = \tan \lambda\,(\cos \varphi n - \mu \tan \lambda) / (\cos \varphi n \tan \lambda + \mu) = (\cos \varphi n - \mu \tan \lambda) / (\cos \varphi n + \mu \cot \lambda)$$

> Efficiency rises with λ (up to about $45°$) and falls with μ; max at $\lambda \approx 45° - \rho'/2$ where $\tan \rho' = \mu / \cos \varphi n$ (ρ' = virtual friction angle).

> **Self-locking (irreversible)**: wheel cannot drive the worm if $\lambda < \rho'$ (i.e. $\tan\lambda < \mu/\cos\varphi_n$). Then $\eta < 50\%$. Used in hoists, lifts; no brake needed - but the worm drive efficiency is low.

> Overall efficiency of the drive also includes bearing and oil churning loss ($\eta_{total} \approx \eta \times 0.95\text{-}0.98$).

Example 3 - worm efficiency

> $z1 = 2$, $m = 5$ mm, $q = 10$ -> $d1 = 50$ mm ; $\tan\lambda = z1/q = 0.2$ -> $\lambda = 11.31°$; $\varphi_n = 20°$, $\mu = 0.05$.

> $\eta = 0.2\ (\cos20° - 0.05 \times 0.2)/(\cos20° \times 0.2 + 0.05) = 0.2 \times 0.9297/(0.1879 + 0.05) = \mathbf{78\%}$. $\rho' = \tan^{-1}(0.05/0.94) = 3.05° < \lambda$ -> not self-locking.

> Single-start worm ($\lambda = 5.71°$) with $\mu = 0.10$: $\eta = 0.1 \times 0.9297/(0.094 + 0.1) = 48\%$; $\rho' = 6.07° > \lambda$ -> **self-locking**.

Stresses in worm gear tooth, design and thermal rating

> The **wheel is weaker** (bronze) so strength is checked on the wheel: **beam strength** $Sb = \sigma b\ Cv\ b\ \pi\ m\ y$, y from Lewis for virtual teeth $z2/\cos^3\lambda$; $Cv = 6/(6 + v2)$ ($v2$ = wheel pitch velocity ≤ 3 m/s) ; σb(phosphor bronze) $\approx 80\text{-}100$ MPa.

> **Wear strength** of wheel: $Sw = d2\ b\ K_w$ (K_w = wear factor, 0.4-1.0 N/mm² by materials from data-book). Condition: $Sb, Sw \geq Fd = Wt2/Cv$.

> Worm: check bending as a beam between bearings (keep deflection small, about 0.001 d1 or less) and torsion/shear; use a thrust bearing for Wa1.

> **Thermal rating**: heat generated $Hg = P\ (1 - \eta)$ (W); heat dissipated $Hd = C\ A\ (t_{oil} - t_{air})$; $A \approx 1.14 \times 10^{-4}\ a^{1.7}$ (m^2, a in mm); $C = 8\text{-}14$ W/m²°C (natural), 20-30 with fan; $t_{oil} \leq 80\text{-}90$ °C. If $Hd < Hg$ -> fins, fan, oil cooler.

> Efficient lubrication (EP-free oil, SAE 140 / ISO VG 220-460 - sulphur additives attack bronze) and run-in are vital.

Design steps for a worm drive

1. Select z1 (from ratio and efficiency/ self-lock need), $z2 = i\ z1$, materials (hardened steel / bronze).
2. Assume centre distance a (or $d1 = a^{0.875}/2.2$) -> q, m, d1, d2; λ, φ_n; v1, Vs, μ.
3. Find $Wt2 = 2\ T2 / d2$; Fd; check wheel Sb and Sw; face width.
4. Efficiency η, power loss, thermal check; forces on bearings; worm shaft design.

V. Additional notes & worked examples

Helical gear - selection of helix angle & thrust direction

β	Effect
10° - 15°	Small thrust; modest smoothness gain; good for heavy loads with limited thrust bearing capacity
15° - 25°	**Usual range** for automobile and industrial single-helical gears
25° - 35°	Very smooth and quiet but high thrust, lower efficiency
30° - 45°	Only for double helical / herringbone gears (thrust cancels)

> **Hand rule for thrust**: for a driving RH helical gear, point the **right hand thumb along the axis**, fingers in the direction of rotation: thumb gives the thrust direction on the gear (use left hand for LH). Driven gear thrust is opposite.

> Length of contact line of helical gear is much longer than a spur gear's face width b: $l = b / \cos \beta_b$ (β_b = base helix angle) - this is the reason for higher load capacity.

> In the automobile gearbox the helix hands on the countershaft gears are chosen so that the **thrusts partly cancel**; the layout is checked for each gear selection.

Example 4 - shaft loads of the helical pair (Example 1)

> $F_t = 2244$ N, $F_r = 869$ N, $F_a = 817$ N, $d_1 = 85.1$ mm. Torque $T = 95{,}500$ N·mm.

> Bending of pinion shaft in vertical plane has an extra moment from the thrust: $M_a = F_a d_1/2 = 817 \times 42.55 =$ **34,760 N·mm**, which adds to the radial-load moment.

> If the pinion is mounted centrally between bearings 100 mm apart: bearing radial reaction (from F_t and F_r) = $\sqrt{(F_t^2 + F_r^2)}/2 = 1203$ N ; thrust 817 N taken by one (fixed) bearing -> choose angular contact / taper roller bearing for that end.

Bevel gears - types

Type	Features	Use
Straight	Teeth straight, radial to apex; simple, sudden engagement; v < 5 m/s	Hand drills, differential (small), low-speed
Spiral	Curved oblique teeth, spiral angle 35°; gradual contact, higher contact ratio, quiet, strong; thrust higher	Automobile rear axle, high-speed drives

Type	Features	Use
Zerol	Curved teeth but spiral angle 0; no extra thrust of spiral	Replace straight bevel when smoother action needed
Hypoid	Axes offset; pinion can be larger and stronger; more sliding, needs EP hypoid oil	**Car final drive**: lowers the propeller shaft (low floor)

> **Differential**: a set of bevel gears (side gears + pinions) which lets the left and right wheels rotate at different speeds in a turn while sharing torque equally; the ring gear / pinion (crown wheel & pinion) gives the final reduction (3:1 to 5:1).

> Bevel mounting: pinion on a rigid shaft with two taper roller bearings spaced apart; preload and set backlash & contact pattern with shims.

Worm gear – force and thermal examples

Example 5 – forces on a worm drive

> $P = 5$ kW, worm speed 1440 rpm, $z1 = 2$, $z2 = 40$, $m = 5$, $q = 10$ ($d1 = 50$, $d2 = 200$ mm), $\eta = 0.78$ (from Example 3). $i = 20$ -> wheel speed 72 rpm.

> $T1 = 9.55 \times 10^6 \times 5/1440 = 33{,}160$ N·mm ; $T2 = T1\, i\, \eta = 33{,}160 \times 20 \times 0.78 = 517{,}300$ N·mm.

> Wheel tangential = worm axial: $Wt2 = Wa1 = 2\,T2/d2 =$ **5173 N** ; worm tangential $Wt1 = 2\,T1/d1 =$ **1326 N** (check: $Wa1 \tan\lambda = 1035$ N frictionless, $1035/1326 = 0.78 = \eta$ ✓).

> Radial $Wr = Wa1 \tan\varphi_n / \cos\lambda = 5173 \times 0.364/0.981 = 1920$ N. Sliding velocity $Vs = \pi \times 50 \times 1440/(60000 \times 0.981) = 3.84$ m/s ($\mu \approx 0.035$-0.05 ✓).

Example 6 – thermal check of the same drive

> Heat generated $Hg = P(1 - \eta) = 5000 \times 0.22 =$ **1100 W**. Casing area $A = 1.14 \times 10^{-4}\, a^{1.7}$ with $a = (50 + 200)/2 = 125$ mm -> $A = 0.42$ m².

> Natural cooling $C = 14$ W/m²°C: $Hd = 14 \times 0.42 \times \Delta t = 5.9\,\Delta t$ -> $\Delta t = 187$ °C ✗ (oil limit about 60 °C rise).

> With fins (A doubled to 0.84 m²) and a fan ($C = 28$): $\Delta t = 1100/(28 \times 0.84) =$ **47 °C** ✓ . The oil temperature stays about 70-80 °C.

Worm wheel – construction

> Wheel rim of bronze (10–15 mm thick) bolted or shrunk on a cast-iron / steel centre to save costly bronze.

> Teeth are cut by a **hob identical to the worm** (so that contact is line contact) – the hob is fed to the correct centre distance; worm & wheel are therefore made as a matched pair.

> Run-in with light load first; after the run-in surfaces polish and efficiency improves.

> Hardness: worm hardness at least 100 BHN above the wheel; ground worm finish ≤ 0.4 µm Ra to cut friction.

VI. Comparison – helical, bevel and worm gears

Point	Helical	Bevel	Worm
Shafts	Parallel (crossed: skew)	Intersecting (90°)	Perpendicular, non-intersecting
Ratio / stage	up to 10	up to 6 (straight)	5 – 100
Efficiency	97–99%	96–98%	40–95%
Thrust	Axial $F_a = F_t \tan\beta$	Axial & radial (cone)	High axial on worm
Noise	Low	Moderate (spiral low)	Very low
Self-locking	No	No	Possible ($\lambda < \rho'$)
Automobile use	Gearbox	Differential, steering	Steering, speedometer

VII. Quick revision & important questions

Formulas to remember

> Helical: $m_t = m_n/\cos\beta$; $a = m_n(z_1 + z_2)/(2\cos\beta)$; $z' = z/\cos^3\beta$; $F_a = F_t \tan\beta$; $F_r = F_t \tan\varphi_n/\cos\beta$; $p_x = \pi m_n/\sin\beta$.

> Bevel: $\tan\delta_1 = z_1/z_2$; $A_0 = d_1/(2\sin\delta_1)$; $b \le A_0/3$; $F_{r1} = F_t \tan\varphi \cos\delta_1$; $F_{a1} = F_t \tan\varphi \sin\delta_1$; $z' = z/\cos\delta$.

> Worm: $\tan\lambda = z_1/q$; $a = m(q + z_2)/2$; $V_s = v/\cos\lambda$; $\eta = (\cos\varphi_n - \mu\tan\lambda)/(\cos\varphi_n + \mu\cot\lambda)$; self-lock $\lambda < \rho'$.

> Strength: $S_b = \sigma_b\, m_n\, b\, Y'$; $S_w = b\, Q\, d_1\, K/\cos^2\beta$ (helical) ; S_b(bevel) has $(1 - b/A_0)$.

1. Explain the kinematics and tooth proportions of a parallel helical gear. (5 M)
2. Derive the forces acting on a helical gear tooth. (5 M)
3. Design a helical gear pair for given power, speed and ratio. (10 M)
4. Write on crossed helical gears: relations and efficiency. (5 M)
5. Derive force components and stresses of a straight bevel gear. (10 M)
6. Design a pair of straight bevel gears for given data. (10 M)
7. Explain worm gearing: kinematics, forces, friction & efficiency, self-locking. (10 M)
8. A worm drive transmits P kW - find efficiency, stresses and the heat to be dissipated. (10 M)